

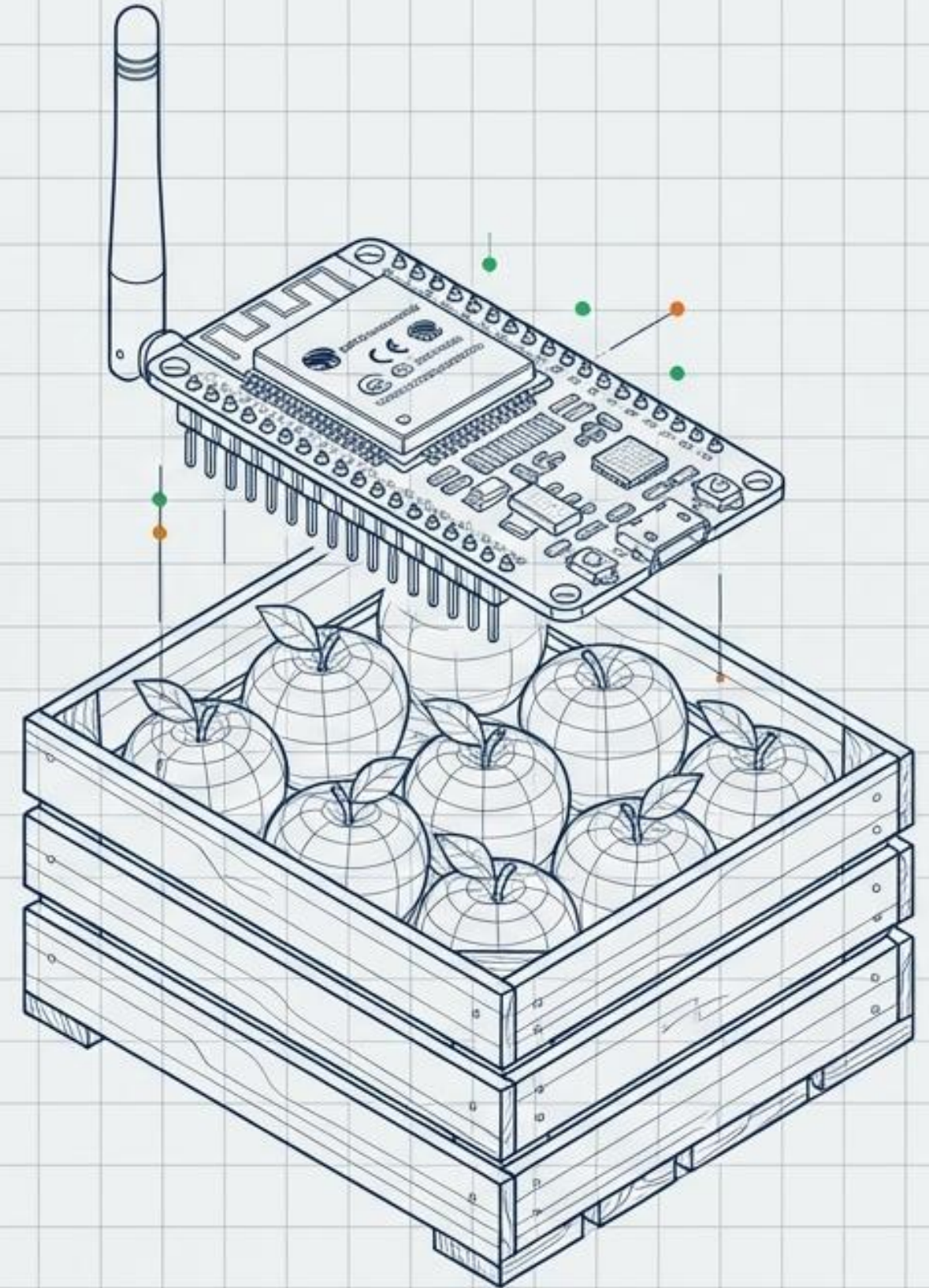
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Intermediate Technical Review: Experiments and Analysis

A Battery-Powered Edge Node for the Early
Detection of Agricultural Spoilage

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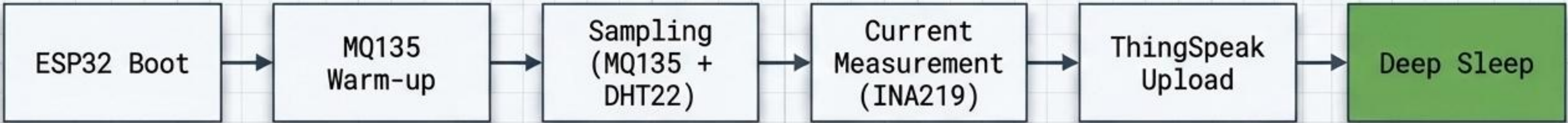
Course: Internet-of-Things
Algorithms and Services 2025-2026



Basic Setup and Operational Sequence

Hardware Roster	
Microcontroller	ESP32 Heltec V3
Environmental Sensors	MQ135 (Gas, GPIO 4) DHT22 (Temp/Humidity, GPIO 5)
Current Sensor	INA219 (Secondary I2C: SDA 6, SCL 7)
Connectivity & Storage	Wi-Fi on ThingSpeak NVS for calibration persistence (R0)

Operating Sequence (Current Duty Cycle: 60 min)



Core Takeaway: The entire sensing pipeline is currently implemented end-to-end.

The Context: Identifying the First Technical Problem

Dataset Adaptation

1. Objective

The reference dataset identified for training classifies fruit spoilage conditions based on CO₂ concentrations.



2. Adaptation

To align with the dataset, we calibrated the MQ135 sensor and implemented a method to estimate the absolute CO₂ value (ppm) from the resistance.



3. Baseline

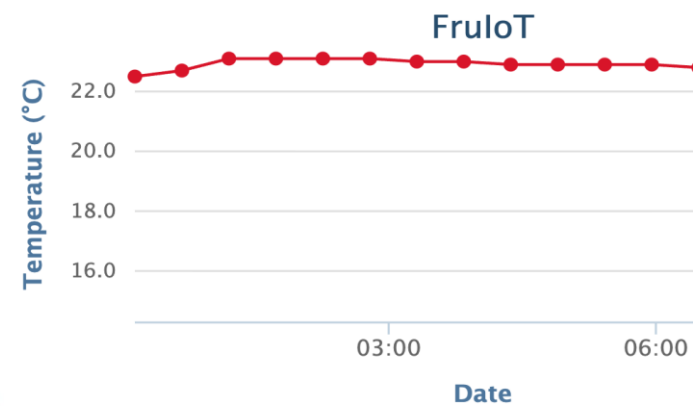
In clean and stable air conditions, the sensor behavior and estimation method proved acceptable.

The Empirical Need:

These initial results prompted us to validate the system in a controlled real-world scenario ("banana-box"), to observe the sensor's reaction to the gases actually emitted by the produce.

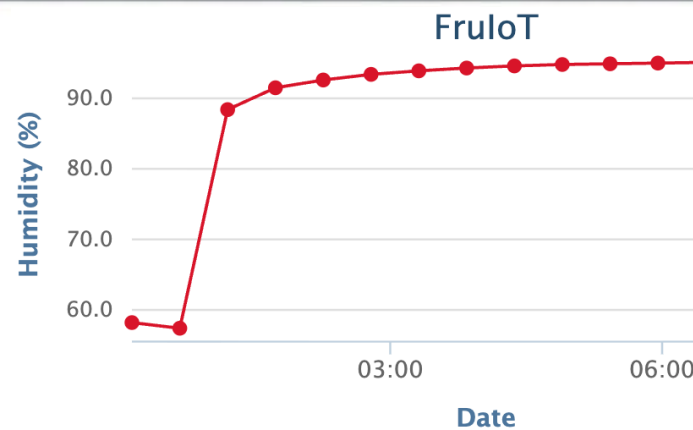
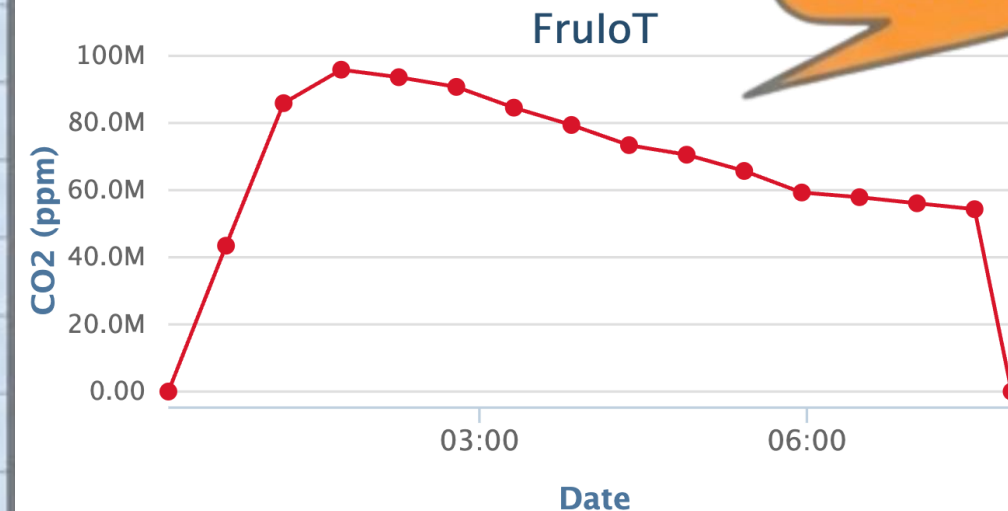
Banana-Box Experiment Results

Temperature



Plausible Data: Stable temperature (~22.5–23.1 °C). Expected sharp increase in humidity in the closed container (~58% -> ~95%).

Humidity

CO₂

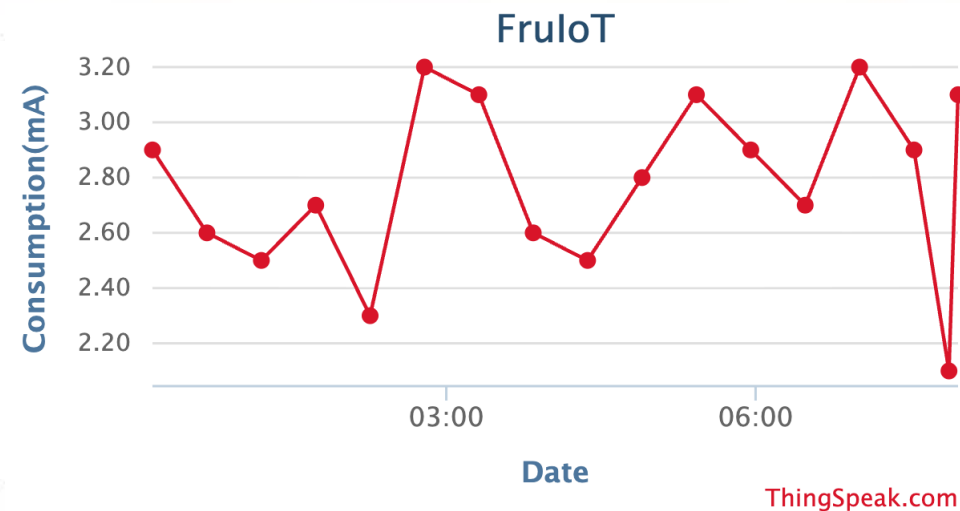
Incorrect Estimate: The CO2 estimation values are off the charts and extremely misleading.

ThingSpeak.com

Problem Analysis: The MQ135 reacts massively to the fruit microenvironment (probably VOCs/ethylene), rendering the absolute estimation of CO₂ unreliable. The main problem is not the sensing hardware, but the reliability of the CO₂ estimation strategy in our scenario.

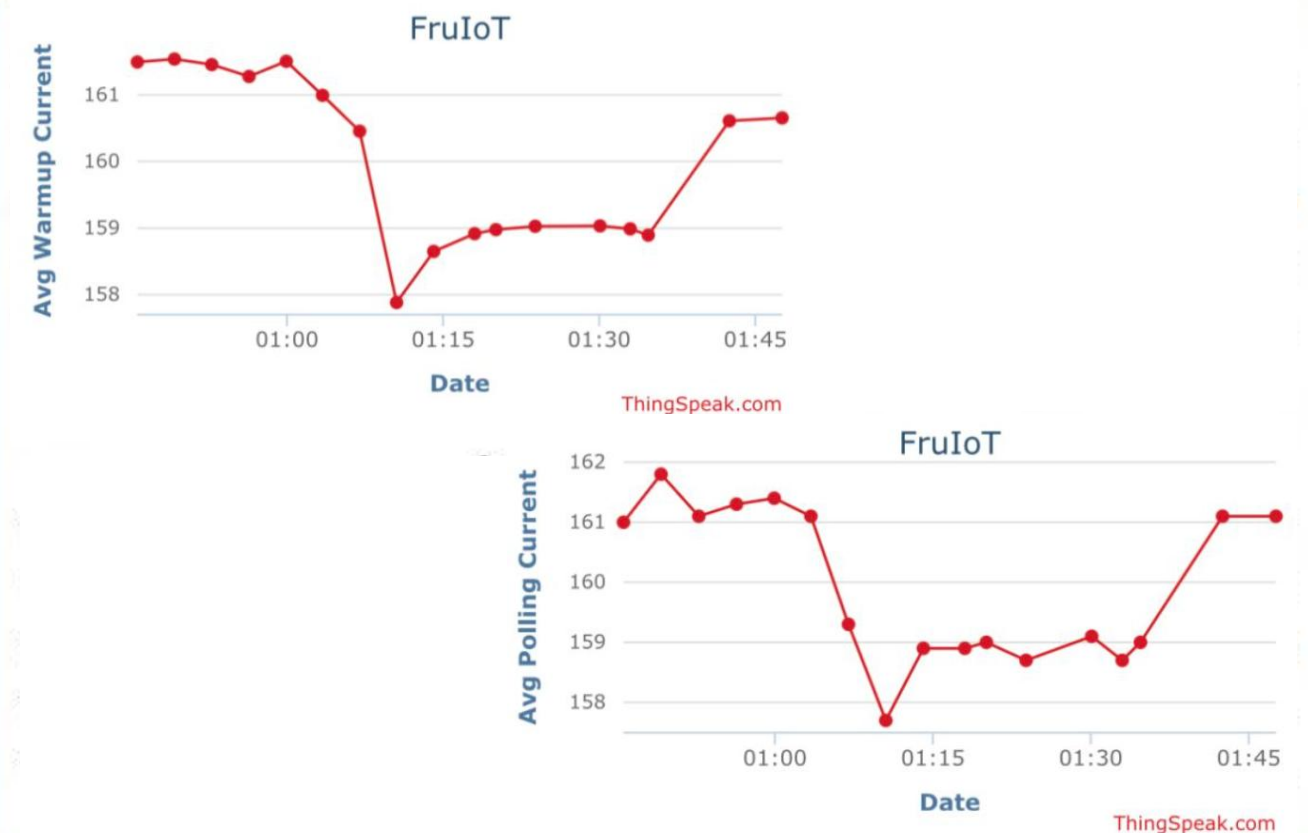
Project Constraints: Duty Cycle and Energy

Microclimate Sampling (DHT22)



Very low power draw (~2-3 mA) in the banana-box scenario.

Gas Sensor Heating (MQ135)



Average measured current (Warm-up & Polling): ~168 mA constant.
Estimate for 98s warm-up: ≈ 4.08 mAh per cycle.
Estimate for 38s warm-up: ≈ 1.33 mAh per cycle.

Ideal battery-life estimate considering only MQ135 warm-up energy (1 cycle/hour, 3000 mAh battery):
~31 days for 90 s preheating and ~94 days for 30 s preheating

Conceptual Map of Next Engineering Choices

